

If a woman plans pregnancy, the benefits and the risks of continuing treatment with dolutegravir should be discussed with the patient.

Pregnancy

There is a small increase of neural tube defects to mothers taking dolutegravir-containing regimens at the time of conception.

The incidence of neural tube defects in the general population ranges from 0.5-1 case per 1,000 live births (0.05-0.1%). Most neural tube defects occur within the first 4 weeks of embryonic development after conception (approximately 6 weeks after the last menstrual period). If a pregnancy is confirmed in the first trimester while on dolutegravir, the benefits and risks of continuing dolutegravir versus switching to another antiretroviral regimen should be discussed with the patient taking the gestational age and the critical time period of neural tube defect development into account.

Data analysed do not indicate an increased risk of major birth defects in women exposed to dolutegravir during pregnancy but are currently insufficient to address the risk of neural tube defects.

No adverse development outcomes, including neural tube defects, were identified. Dolutegravir was shown to cross the placenta in animals.

More than 1000 outcomes from exposure during second and third trimester of pregnancy indicate no evidence of increased risk of foeto/neonatal toxicity. Dolutegravir may be used during the second and third trimester of pregnancy when the expected benefit justifies the potential risk to the foetus.

Breast-feeding

Dolutegravir is excreted in human milk in small amounts. There is insufficient information on the effects of dolutegravir in neonates/infants.

It is recommended that HIV infected women do not breast-feed their infants under any circumstances in order to avoid transmission of HIV.

Fertility

There are no data on the effects of dolutegravir on human male or female fertility. Animal data indicate no effects of dolutegravir on male or female fertility

4.7 Effects on ability to drive and use machines

Patients should be informed that dizziness has been reported during treatment with dolutegravir. The clinical status of the patient and the adverse reaction profile of dolutegravir should be borne in mind when considering the patient's ability to drive or operate machinery.

4.8 Undesirable effects

Summary of the safety profile

The most severe adverse reaction, seen in an individual patient, was a hypersensitivity reaction that included rash and severe liver effects (see section 4.4). The most commonly seen treatment emergent adverse reactions were nausea, diarrhoea and headache.

Tabulated list of adverse reactions

The adverse reactions considered at least possibly related to dolutegravir are listed by body system, organ class and absolute frequency. Frequencies are defined as very common, common, uncommon, rare, very rare. Table 2 Adverse Reactions

Immune system disorders	Uncommon	Hypersensitivity (see section 4.4)
	Common	Immune Reconstitution Syndrome (see section 4.4)**
Psychiatric disorders	Common	Insomnia
	Common	Abnormal dreams
	Common	Depression
	Common	Anxiety
	Uncommon	Suicidal ideation*, suicide attempt* *particularly in patients with a pre-existing history of depression or psychiatric illness.
Nervous disorders	Very common	Headache
	Common	Dizziness
Gastrointestinal disorders	Very common	Nausea
	Very common	Diarrhoea
	Common	Vomiting
	Common	Flatulence
	Common	Upper abdominal pain
	Common	Abdominal pain
	Common	Abdominal discomfort
Hepatobiliary disorders	Common	Alanine aminotransferase (ALT) and/or Aspartate aminotransferase (AST) elevations
	Uncommon	Hepatitis
	Rare	Acute hepatic failure, increased bilirubin***
Skin and subcutaneous tissue disorders	Common	Rash
	Common	Pruritus
Musculoskeletal and connective tissue disorders	Uncommon	Arthralgia
	Uncommon	Myalgia
General disorders and administration site conditions	Common	Fatigue
Investigations	Common	Creatine phosphokinase (CPK) elevations

**See below under Description of selected adverse reactions.

***In combination with increased transaminases

Description of selected adverse reactions

Changes in laboratory biochemistries

Increases in serum creatinine occurred within the first week of treatment with dolutegravir and remained stable through 48 weeks. A mean change from baseline of 9.96 µmol/L was observed after 48 weeks of treatment. Creatinine increases were comparable by various background regimens. These changes are not considered to be clinically relevant since they do not reflect a change in glomerular filtration rate.

Co-infection with Hepatitis B or C

Overall, the safety profile in patients co-infected with hepatitis B and/or C was similar to that observed in patients without hepatitis B or C co-infection, although the rates of AST and ALT abnormalities were higher in the subgroup with hepatitis B and/or C co-infection for all treatment groups. Liver chemistry elevations consistent with immune reconstitution syndrome were observed in some patients with hepatitis B and/or C co-infection at the start of dolutegravir therapy, particularly in those whose anti-hepatitis B therapy was withdrawn (see section 4.4).

Immune reactivation syndrome

In HIV-infected patients with severe immune deficiency at the time of initiation of combination antiretroviral therapy (CART), an inflammatory reaction to asymptomatic or residual opportunistic infections may arise. Autoimmune disorders (such as Graves' disease and autoimmune hepatitis) have also been reported; however, the reported time to onset is more variable and these events can occur many months after initiation of treatment (see section 4.4).

Metabolic parameters

Weight and levels of blood lipids and glucose may increase during antiretroviral therapy (see section 4.4).

Paediatric population

Based on available data, children and adolescents (aged 4 weeks and above, to less than 18 years, and weighing at least 3 kg) who received the recommended doses of film-coated tablets or dispersible tablets once daily, there were no additional types of adverse reactions beyond those observed in the adult population.

4.9 Overdose

There is currently limited experience with overdosage in dolutegravir.

Limited experience of single higher doses (up to 250 mg) revealed no specific symptoms or signs, apart from those listed as adverse reactions.

Further management should be as clinically indicated or as recommended by the national poisons centre, where available. There is no specific treatment for an overdose of dolutegravir. If overdose occurs, the patient should be treated supportively with appropriate monitoring, as necessary. As dolutegravir is highly bound to plasma proteins, it is unlikely that it will be significantly removed by dialysis.

5. Pharmacological properties

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Antivirals for systemic use, other antivirals, ATC code: J05AJ03

Mechanism of action

Dolutegravir inhibits HIV integrase by binding to the integrase active site and blocking the strand transfer step of retroviral Deoxyribonucleic acid (DNA) integration which is essential for the HIV replication cycle.

Pharmacodynamic effects

Antiviral activity in cell culture

The IC₅₀ for dolutegravir in various labstrains using PBMC was 0.5 nM, and when using MT-4 cells it ranged from 0.7-2 nM. Similar IC₅₀ were seen for clinical isolates without any major difference between subtypes; in a panel of 24 HIV-1 isolates of clades A, B, C, D, E, F and G and group Q the mean IC₅₀ value was 0.2 nM (range 0.02-2.14). The mean IC₅₀ for 3 HIV-2 isolates was 0.16 nM (range 0.09-0.61).

Antiviral activity in combination with other antiviral agents

No antagonistic effects were seen with dolutegravir and other antiretrovirals tested: stavudine, abacavir, efavirenz, nevirapine, lopinavir, amprenavir, enfuvirtide, maraviroc and raltegravir. In addition, no antagonistic effects were seen for dolutegravir and adefovir, and ribavirin had no apparent effect on dolutegravir activity.

Effect of human serum

In 100% human serum, the mean protein fold shift was 75 fold, resulting in protein adjusted IC90 of 0.064

µg/mL Resistance.

Mutations selected appeared slowly using the lab-strain HIV-1 IIIB during passage, with substitutions at positions S153Y and F, resulting in a maximal fold change in susceptibility of 4.

Mutation R263K was reported from ART experienced, INI naive individual patients with subtypes B and C, but without effects on dolutegravir susceptibility. G118R lowers the susceptibility to dolutegravir in site directed mutants (FC 10), but was not detected in patients receiving dolutegravir.

Primary mutations for raltegravir/elvitegravir (Q148H/R/K, N155H, Y143R/H/C, E92Q and T66I) do not affect the susceptibility of dolutegravir as single mutations. When mutations listed as secondary integrase inhibitor associated mutations (for raltegravir/elvitegravir) are added to these primary mutations with site directed mutants, dolutegravir susceptibility is still unchanged (FC <2 vs wild type virus), except in the case of Q148-mutations, where a FC of 5-10 or higher is seen with combinations of certain secondary mutations. With strain NL432, starting with site directed mutants harbouring N155H or E92Q, no further selection of resistance was seen (FC unchanged around 1). In contrast, starting with mutants harbouring mutation Q148H (FC 1), a variety of secondary mutations were seen with a consequent increase of FC to values >10.

A clinically relevant phenotypic cut-off value (FC vs wild type virus) has not been determined; genotypic resistance was a better predictor for outcome.

Dolutegravir has a less than or equal to 10 FC against 94% of isolates from raltegravir experienced patients.

Resistance

In previously untreated patients receiving dolutegravir + 2 NRTIs, no development of resistance to the integrase class, or to the NRTI class was seen. In previously untreated patients receiving dolutegravir + lamivudine through week 96, no development of resistance to the integrase class, or to the NRTI class was seen.

In patients with prior failed therapies, but naïve to the integrase class, integrase inhibitor substitutions were observed in patients treated with dolutegravir, which was given in combination with an selected background regimen (BR).

In the presence of integrase class-resistance, treatment emergent integrase resistance typically appeared in patients with a history of the Q148-mutation (baseline or historic). Treatment-emergent mutations or mixtures of mutations observed were L74I, N155H.

In paediatric patients with prior failed therapies, but naïve to the integrase class, the integrase inhibitor substitution G118R was observed in patients treated with dolutegravir, given in combination with an selected background regimen. Dolutegravir FC (fold change as compared to wildtype virus) ranged from 6 to 25-fold.

Effects on electrocardiogram

No relevant effects were seen on the QTc interval, with doses exceeding the clinical dose by approximately three fold.

5.2 Pharmacokinetic properties

Dolutegravir pharmacokinetics are similar between healthy and HIV-infected patients. The PK variability of dolutegravir is low to moderate. Between-patient CVb% for AUC and C_{max} ranged from ~20 to 40% and C_{min} from 30 to 65%. The between-patient PK variability of dolutegravir was higher in HIV-infected patients than healthy patients. Within-patient variability (CVw%) is lower than between-patient variability.

Film-coated tablets and dispersible tablets do not have the same bioavailability. The relative bioavailability of dispersible tablets is approximately 1.6-fold higher as compared to film-coated tablets. Thus, a 50 mg dolutegravir dose administered as film-coated tablet(s) will have similar exposure to a 30 mg dolutegravir dose administered as six 5 mg dispersible tablets. Similarly, a 40 mg dolutegravir dose administered as four 10 mg film-coated tablets will provide comparable exposure to a 25 mg dolutegravir dose administered as five 5 mg dispersible tablets.

Absorption

Dolutegravir is rapidly absorbed following oral administration, with median T_{max} at 2 to 3 hours post dose for tablet formulation.

Food increased the extent and slowed the rate of absorption of dolutegravir. Bioavailability of dolutegravir depends on meal content: low, moderate, and high fat meals increased dolutegravir AUC_{0-∞} by 33%, 41%, and 66%, increased C_{max} by 46%, 52%, and 67%, prolonged T_{max} to 3, 4, and 5 hours from 2 hours under fasted conditions, respectively for the film-coated tablet. These increases may be clinically relevant in the presence of certain integrase class resistance. Therefore, MYLTEGA is recommended to be taken with food by patients infected with HIV with integrase class resistance (see section 4.2).

The absolute bioavailability of dolutegravir has not been established.

Distribution

Dolutegravir is highly bound (>99%) to human plasma proteins based on data. The apparent volume of distribution is 17 L to 20 L in HIV-infected patients. Binding of dolutegravir to plasma proteins is independent of dolutegravir concentration. Total blood and plasma drug-related radioactivity concentration ratios averaged between 0.441 to 0.535, indicating minimal association of radioactivity with blood cellular components. The unbound fraction of dolutegravir in plasma is increased at low levels of serum albumin (<35 g/L) as seen in patients with moderate hepatic impairment.

Dolutegravir is present in cerebrospinal fluid (CSF). In treatment-naïve patients on a stable dolutegravir plus abacavir/lamivudine regimen, dolutegravir concentration in CSF averaged 18 ng/mL (comparable to unbound plasma concentration, and above the IC₅₀).

Dolutegravir is present in the female and male genital tract. AUC in cervicovaginal fluid, cervical tissue and vaginal tissue were 6-10% of those in corresponding plasma at steady state. AUC in semen was 7% and 17% in rectal tissue of those in corresponding plasma at steady state.

Biotransformation

Dolutegravir is primarily metabolized through glucuronidation via UGT1A1 with a minor CYP3A component. Dolutegravir is the plasma predominant circulating component. In general, dolutegravir exhibited nonlinear pharmacokinetics with less than 1% of the dose). Fifty-three percent of total oral dose is excreted unchanged in the faeces. It is unknown if all or part of this is due to unabsorbed active substance or biliary excretion of the glucuronide conjugate, which can be further degraded to form the parent compound in the gut lumen. Thirty-two percent of the total oral dose is excreted in the urine, represented by either glucuronide of dolutegravir (18.9% of total dose), N-dealkylation metabolite (3.6% of total dose), and a metabolite formed by oxidation at the benzylic carbon (3.0% of total dose).

Drug interactions

Dolutegravir demonstrated no direct, or weak inhibition (IC₅₀>50 µM) of the enzymes cytochrome P₄₅₀ (CYP)1A2, CYP2A6, CYP2B6, CYP2C8, CYP2C9, CYP2C19, CYP2D6 CYP3A, uridine diphosphate glucuronosyl transferase (UGT)1A1 or UGT2B7, or the transporters Pgp, BCRP, BSEP, OATP1B1, OATP1B3, OCT1, MATE2-K, MRP2 or MRP4. Dolutegravir did not induce CYP1A2, CYP2B6 or CYP3A4. Based on this data, dolutegravir is not expected to affect the pharmacokinetics of medicinal products that are substrates of major enzymes or transporters (see section 4.5).

Dolutegravir was not a substrate of human OATP 1B1, OATP 1B3 or OCT 1.

Elimination

Dolutegravir has a terminal half-life of ~14 hours. The apparent oral clearance (CL/F) is approximately 1L/hr in HIV-infected patients based on a population pharmacokinetic analysis.

Linearity/non-linearity

The linearity of dolutegravir pharmacokinetics is dependent on dose and formulation. Following oral administration of film-coated tablet formulations, in general, dolutegravir exhibited nonlinear pharmacokinetics with less than dose-proportional increases in plasma exposure from 2 to 100 mg; however increase in dolutegravir exposure appears dose proportional from 25 mg to 50 mg for the film-coated tablet formulation. With 50 mg film-coated tablet twice daily, the exposure over 24 hours was approximately doubled compared to 50 mg film-coated tablet once daily.

Pharmacokinetic/pharmacodynamic relationship(s)

In HIV-1-infected patients treated with dolutegravir monotherapy (ING111521), it demonstrated rapid and dose-dependent antiviral activity, with mean decline in HIV-1 RNA of 2.5 log₁₀ at day 11 for 50 mg dose. This antiviral response was maintained for 3 to 4 days after the last dose in the 50 mg film-coated tablet group.

PK/PD modelling in integrase resistant patients suggest that increasing the dose from 50 mg film-coated tablet twice daily to 100 mg film-coated tablet twice daily may increase the effectiveness of dolutegravir in patients with integrase resistance and limited treatment options due to advanced multi class resistance. The proportion of responders (HIV-1 RNA <50 c/mL) at week 24 was predicted to increase around 4-18% in the patients with Q148 + ≥2 secondary mutations from G140A/C/S, E138A/K/T, L74I. This high dose may be considered in the presence of the Q148 + ≥2 secondary mutations from G140A/C/S, E138A/K/T, L74I in patients with overall limited treatment options due to advanced multi class resistance. There is no clinical data on the safety or efficacy of the 100 mg film-coated tablet twice daily dose. Co-treatment with atazanavir increases the exposure of dolutegravir markedly, and should not be used in combination with this high dose, since safety with the resulting dolutegravir exposure has not been established.

Special patient populations

Children

Steady state simulated plasma exposure at once daily weight band doses in HIV-1 infected infants, children and adolescents aged ≥ 4 weeks to < 18 years is summarized in Table 3.

Table 3 Summary of Simulated Dolutegravir PK Parameters at Once Daily Doses by Weight Band in Paediatric HIV-1 Infected Patients

Weight (kg)	Band	Dolutegravir Dosage Form*	Once Daily Dose (mg)	PK Parameter C _{max} (µg/mL)	AUC _{0-24h} (µg* ^h /mL)	C _{24h} (ng/mL)
3 to <6		DT	5	4.02	49.4	1070
6 to <10 ^a		DT	10	5.90	67.4	1240
6 to <10 ^b		DT	15	6.67	68.4	964
10 to <14		DT	20	6.61	63.1	719
14 to <20		DT FCT	25 40	7.17 6.96	69.5 72.6	824 972
20 to <25		DT FCT	30 50	7.37 7.43	72.0 78.6	881 1080
25 to <30		FCT	50	6.74	71.4	997
30 to <35		FCT	50	6.20	66.6	944

≥ 35	FCT	50	4.93	54.0	814	
Target: Geometric Mean				46	995	
DT=dispersible tablet FCT=film-coated tablet a. The bioavailability of dolutegravir DT is ~ 1.6-fold dolutegravir FCT. b. <6 months of age c. ≥6 months of age						
Steady state simulated plasma exposure at alternative twice daily weight band doses are summarized in table 4.						
Table 4 Summary of Simulated Dolutegravir PK Parameters at Alternative Twice Daily Doses by Weight Band in Paediatric HIV-1 Infected Patients						
Weight (kg)	Band	Dolutegravir Dosage Form*	Twice Daily Dose (mg)	PK Parameter		
				Cmax (µg/mL)	AUC0-12h (µg·h/mL)	C12h (ng/mL)
6 to <10 ^a		DT	5	4.28	31.6	1760
6 to <10 ^c		DT	10	6.19	43.6	2190
10 to <14		DT	10	4.40	30.0	1400
14 to <20		DT FCT	15 20	5.78 4.98	39.6 35.9	1890 1840
20 to <25		DT FCT	15 25	5.01 5.38	34.7 39.2	1690 2040
25 to <30		DT FCT	15 25	4.57 4.93	32.0 35.9	1580 1910
30 to <35		FCT	25	4.54	33.3	1770
≥35		FCT	25	3.59	26.8	1470
DT=dispersible tablet FCT=film-coated tablet a. The bioavailability of dolutegravir DT is ~ 1.6-fold dolutegravir FCT. b. <6 months of age c. ≥6 months of age						

Elderly

In HIV-1 infected adults showed that there was no clinically relevant effect of age on dolutegravir exposure. Pharmacokinetic data for dolutegravir in patients >65 years of age are limited.

Renal impairment

Renal clearance of unchanged active substance is a minor pathway of elimination for dolutegravir. The exposure to dolutegravir was decreased by approximately 40% in patients with severe renal impairment. The mechanism for the decrease is unknown. No dosage adjustment is considered necessary for patients with renal impairment. There is no data of use of Dolutegravir in patients on dialysis.

Hepatic impairment

Dolutegravir is primarily metabolized and eliminated by the liver. While the total dolutegravir concentration in plasma was similar, a 1.5- to 2-fold increase in unbound exposure to dolutegravir was observed in patients with moderate hepatic impairment compared to healthy patients. No dosage adjustment is considered necessary for patients with mild to moderate hepatic impairment. There is no data on effect of severe hepatic impairment on the pharmacokinetics of Dolutegravir.

Polymorphisms in drug metabolising enzymes

There is no evidence that common polymorphisms in drug metabolising enzymes alter dolutegravir pharmacokinetics to a clinically meaningful extent. Patients with UGT1A1 genotypes conferring poor dolutegravir metabolism had a 32% lower clearance of dolutegravir and 46% higher AUC via UGT1A1.

Gender

There is no clinically relevant effect of gender on the exposure of dolutegravir.

Race

There is no clinically relevant effect of race on the exposure of dolutegravir. The pharmacokinetics of dolutegravir following single dose oral administration to Japanese patients appear similar to observed parameters in Western (US) patients

Co-infection with Hepatitis B or C

Population pharmacokinetic analysis indicated that hepatitis C virus co-infection had no clinically relevant effect on the exposure to dolutegravir. There are limited data on patients with hepatitis B co-infection.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Core tablet: mannitol, povidone, sodium starch glycolate, microcrystalline cellulose and sodium stearyl fumarate.

Film coating (Opadry II brown 85F565142): polyvinyl alcohol, macrogol/polyethylene glycol, talc, titanium dioxide, red iron oxide and black iron oxide.

6.2 Incompatibilities

Not applicable

6.3 Shelf life

36 months

6.4 Special precautions for storage

Do not store above 30°C. Store in the original container.

6.5 Nature and contents of container

Tablets are packed in round blue opaque HDPE bottle with blue opaque polypropylene cap with wad containing aluminium induction sealing liner.

Pack sizes: 30 tablets.

6.6 Instruction for use, handling and disposal

No special requirements

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

7. Manufacturer

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10. DATE OF REVISION OF THE TEXT

April 2022

The Product has been produced under a licence from the Medicines Patent Pool.

The Product is not authorized for supply to the Private Market

Any other use is not authorized

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